

Infrastructure and Indirect Benefits in Cities: Evidence from City Level Data

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The Quito subway began operations in 2023.

- 148k passengers/day in 2024 = 54m for the year City of Quito (2025).
- 2.0b for construction. At 5% interest, debt service is 100m/year or 1.85/rider World Bank (2018).
- 2024 operating expenses 43m or 0.80/rider Municipio del Distrito Metropolitano de Quito (2025).
- Total cost is 2.65/rider. Fare is 0.45. Quito lost 2.20 for each of 54m riders in 2024. 119m in total or 59.00/Quito resident.
- Does this subsidy make sense? We need to find 2.20 of benefits to *non-riders* to make up the difference, or else justify a transfer to riders.
- Infrastructure projects often face similar cost benefit calculations. What does city level data say about the indirect benefits of highways, subways, BRTs, and water and sewer infrastructure?

Within (not between) City Roads and Highways

- Cause suburbanization Baum-Snow (2007); Baum-Snow et al. (2017).
- Have economically small effects on population and employment growth e.g., Duranton and Turner (2012), Frye (2024), Baum-Snow et al. (2017), Baum-Snow (2020). Also affect productivity Holl (2016), Baum-Snow (2020), Ghani et al. (2016).
- Shift production towards tradable goods Duranton et al. (2014).
- Do not reduce congestion Duranton and Turner (2011), but increase speed Currier et al. (2023), Akbar et al. (2023).
- Effect on pollution *exposure* is ambiguous.
- Effect on CO2 is positive Duranton and Turner (2011)

Indirect effects on total travel, CO2, and suburbanization are obviously large. The indirect effects on population growth and employment growth look small. It's hard to tell how big the effects on firm productivity are.

Subways

- Decentralize *lights* in cities in much the same way that roads decentralize populations Gonzalez-Navarro and Turner (2018).
- Probably do not reduce traffic Duranton and Turner (2011), except on parallel roads Gu et al. (2021).
- Cause economically important reductions in particulates, worth about 1/3 of construction costs in a highly polluted city Gendron-Carrier et al. (2022).
- Shift employment to formal sector in Mexico City. Zárate (2022)
- Effects on CO2 are indeterminant.

Indirect effects on particulates can be important in polluted cities. Effects on population growth and employment growth are probably about zero. We know less about subways than roads.

BRTs

- Pollution effect is about 10% of subways, maybe more near CBD Gendron-Carrier et al. (2026)
- Concentrates lights in the center Gendron-Carrier et al. (2026). *This appears to be different from subways and roads.* This is a challenge for theory.
- Extrapolating from subways, BRTs do not reduce traffic Duranton and Turner (2011), except possibly on parallel roads Gu et al. (2021).
- Transmilenio increases Bogota real income $\approx 2\%$ Tsivanidis (2019). Commuters sort close to BRT in Dar es Salam Balboni et al. (2025).

Model based evidence for increased income about double the value of time saved in Bogota. Small increase to total lights on average. No effect on particulates on average, Probably no effect on total employment, population, or CO2. We know less about BRTs than subways.

Water and Sewer

- Water and sewer infrastructure decreases infant mortality, crude death rate, and morbidity in 19th century US and Europe, probably dramatically Anderson et al. (2018), Alsan and Goldin (2019), Ferrie and Troesken (2008), Kesztenbaum and Rosenthal (2017).
- Large effects of improved water in Latin America on infant mortality Galiani et al. (2005) Bhalotra et al. (2021). Less from sewers Gamper-Rabindran et al. (2010).
- Water supplies increase leisure time (but not childhood illness) Devoto et al. (2012). Reliable drinking water decreases childhood illness and increases girls' schooling Ashraf et al. (2017).
- Water and sewer doubles land values Coury et al. (2024). Sewers increase population density McCulloch et al. (2025).

Water and sewer infrastructure can have enormous effects on health and the development of cities. Comparisons to transportation infrastructure are difficult, but favor water and sewer.

Conclusion I

- Transportation (and other) infrastructure often requires large subsidies. City level data tells us something about the magnitudes of the indirect benefits that can be used to justify these subsidies.
- For example, Quito's subway has attracted riders. But, can we find indirect benefits to non-riders to justify a 75% subsidy? Are these benefits external or private? Quito has clean air, so improvements to air quality are likely minor. Other indirect benefits are more speculative.
- BRTs are similar to subways, but concentrate economic activity in the center and have a smaller effect on pollution.

Conclusion II

- Roads decentralize population and employment, likely increase pollution, but may reduce exposure. They probably do not reduce congestion, but likely increase travel speed. Indirect benefits and costs are probably larger than for subways and BRTs, e.g., Baum-Snow (2020).
- Water and sewer infrastructure appears to have much larger indirect effects than transportation infrastructure, but no cost benefit analyses are available.

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